15 **(a)** Show that $\sum_{r=1}^{\infty} 0.99^{r-1} \times 0.01 = 1$. **[3]**

Kofi is a very good table tennis player. Layla is determined to beat him.

Every week they play one match of table tennis against each other. They will stop playing when Layla wins the match for the first time.

X is the discrete random variable "the number of matches they play in total".

Kofi models the situation using the probability function

$$P(X = r) = 0.99^{r-1} \times 0.01 \quad r = 1, 2, 3, 4, \ldots$$

Kofi states that he is 95% certain that Layla will not beat him within 6 years.

(b) Determine whether Kofi's statement is consistent with his model. **[3]**

In between matches, Layla practises, but Kofi does not.

(c) Explain why Layla might disagree with Kofi's model. **[1]**

Layla models the situation using the probability function

$$P(X = r) = kr^2 \qquad r = 1, 2, 3, 4, 5, 6, 7, 8.$$

(d) Explain how Layla's model takes into account the fact that she practises between matches, but Kofi's does not. **[1]**

Layla states that she is 95% certain that she will beat Kofi within the first 6 matches.

(e) Determine whether Layla's statement is consistent with her model. **[3]**